

SkilPrep — Project Synopsis & Developer Report

Synopsis

SkilPrep is a multi-discipline challenge and learning platform that supports coding, mathematics, science, logic, and general-knowledge exercises. It provides problem authoring, in-browser code editing and execution, automatic and manual evaluation, community tests, shared resources, and instructor (admin) workflows for review and feedback.

Architecture

- **Client:** React + Vite SPA (dev server at <http://localhost:5173/>).
- **Server/API:** Node.js + Express providing REST endpoints under `server/src`.
- **Database:** MongoDB with Mongoose models for Users, Problems, Submissions, Communities, Tests, Resources.
- **Sandbox & Evaluators:** Modular evaluators in `server/src/evaluators` and a sandbox for safe code execution.
- **File uploads:** handled via `multer` and stored in `uploads/`.

Tech Stack & Rationale

- **React + Vite:** fast dev feedback and modern frontend toolchain.
- **Node + Express:** lightweight REST API framework and large ecosystem.
- **MongoDB + Mongoose:** flexible data models for documents like problems and submissions.
- **Monaco Editor:** production-ready in-browser code editor for coding problems.
- **JWT + bcrypt:** stateless auth plus secure password storage.
- **Evaluators:** separate scoring logic per problem type makes system extensible and safer.

Development Workflow

1. Configure environment: create `server/.env` and set `MONGO_URI`.
2. Install dependencies for `root`, `server`, and `client`.
3. Run seed data via `npm run seed` to populate demo users, problems, communities, and tests.
4. Start both dev servers from project root: `npm run dev` (uses concurrently).
5. Server changes are auto-reloaded by `nodemon`; client changes are hot-reloaded by Vite.

Seed / Demo Dataset Summary

The seed script at `server/seed/runSeed.js` populates a presentation dataset and writes `server/seed/presentation-dataset-report.json` describing counts and credentials.

Item	Count / Notes
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Users	6 (admin + 5 demo learners)
Problems	15 covering coding, math, science, logic, GK
Submissions	14 (mixed accepted & wrong answers)
Community	1 community <i>skilprep-founders-circle</i> (private) with a published test and resources

Admin/demo credentials (from seed)

```
admin / admin@skilprep.local password: admin123
alice_dev / alice@skilprep.local password: demo1234
bob_math / bob@skilprep.local password: demo1234
charlie_science / charlie@skilprep.local password: demo1234
diana_logic / diana@skilprep.local password: demo1234
eva_rookie / eva@skilprep.local password: demo1234
```

Where to look in the code

- **Server entry:** `server/src/server.js` and `server/src/app.js`.
- **Routes:** `server/src/routes/` (auth, problems, community, tests, submissions).
- **Models:** `server/src/models/` (User, Problem, Submission, Community, etc.).
- **Evaluators:** `server/src/evaluators/` (MathEvaluator, McqEvaluator, ShortAnswerEvaluator).
- **Seed script:** `server/seed/runSeed.js` — generates demo data and `presentation-dataset-report.json`.
- **Client:** `client/src/pages` and `client/src/components`.

Next steps & Recommendations

- Secure production environment: use a managed MongoDB, HTTPS, and rotate demo credentials before sharing externally.
- Harden sandbox execution and resource limits for production workloads.
- Add automated tests for evaluators and API endpoints.
- Consider CI pipeline to run linting, tests, and seed verification.